

what is cloud computing?

Cloud computing is on demand delivery of compute power, database storage, applications and other IT resources.

* Through cloud services platform pay as you go pricing.

→ They have big server, small server etc.

* You can provision exactly right type and size of computing resources you need

* You can access as many resources as you need, almost instantly

* Amazon web services owns and maintains network connected hardware required for these application services, while you provision and use what you need via web application.

Ex- take office, instead of building own datacenter we have the cloud. We are going to have servers as many as we need. We are just going to pay exactly what we're using.

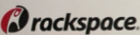
You've been using some Cloud services

		
Gmail	Dropbox	Netflix
• E-mail cloud service • Pay for ONLY your emails stored (no infrastructure, etc.)	• Cloud Storage Service • Originally built on AWS	• Built on AWS • Video on Demand

The Deployment Models of the cloud:-

The Deployment Models of the Cloud

Private Cloud: • Cloud services used by a single organization, not exposed to the public. • Complete control • Security for sensitive applications • Meet specific business needs	Public Cloud: • Cloud resources owned and operated by a third-party cloud service provider delivered over the Internet. • Six Advantages of Cloud Computing	Hybrid Cloud: • Keep some servers on premises and extend some capabilities to the Cloud • Control over sensitive assets in your private infrastructure • Flexibility and cost-effectiveness of the public cloud
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The Five Characteristics of Cloud Computing

- **On-demand self service:**
 - Users can provision resources and use them without human interaction from the service provider
- **Broad network access:**
 - Resources available over the network and can be accessed by diverse client platforms
- **Multi-tenancy and resource pooling:**
 - Multiple customers can share the same infrastructure and applications with security and privacy
 - Multiple customers are serviced from the same physical resources
- **Rapid elasticity and scalability:**
 - Automatically and quickly acquire and dispose resources when needed
 - Quickly and easily scale based on demand
- **Measured service:**
 - Usage is measured, users pay correctly for what they have used

Six Advantages of Cloud Computing

- Trade capital expense (CAPEX) for operational expense (OPEX)
 - Pay On-Demand: don't own hardware
 - Reduced Total Cost of Ownership (TCO) & Operational Expense (OPEX)
- Benefit from massive economies of scale
 - Prices are reduced as AWS is more efficient due to large scale
- Stop guessing capacity
 - Scale based on actual measured usage
- Increase speed and agility
- Stop spending money running and maintaining data centers

Problems solved by cloud :-

- * **Flexibility** : — change resource types when needed .
 - * **Cost Effectiveness** : — pay as you go , for what you use .
 - * **Elasticity** : — ability to scale out and scale in when needed .
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 - * **Scalability** :- Accommodate larger loads by making hardware stronger or adding additional nodes .
 - * **High availability and fault tolerance** :- build across data centers .

* **Agility** : - rapidly develop, test and launch software Applications

Types of cloud computing : -

1) **Infrastructure as a Service (IaaS)** : -

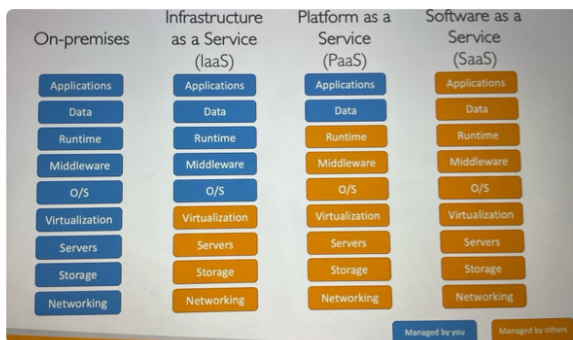
- * provide building blocks for cloud IT * provides Networking, Computers data Storage Space
- * Highest level of flexibility * Easy parallel with traditional on premises IT

2) **Platform as a Service (PaaS)** : -

- * Removes need for your organization to manage underlying Infrastructure.
- * Focus on deployment and management of your applications.

3) **Software as a Service (SaaS)** : -

Completed product that is run and managed by Service Provider.



Example of Cloud Computing Types

- **Infrastructure as a Service:**
 - Amazon EC2 (on AWS)
 - GCP, Azure, Rackspace, Digital Ocean, Linode
- **Platform as a Service:**
 - Elastic Beanstalk (on AWS)
 - Heroku, Google App Engine (GCP), Windows Azure (Microsoft)
- **Software as a Service:**
 - Many AWS services (ex: Rekognition for Machine Learning)

Pricing of the Cloud – Quick Overview

- AWS has 3 pricing fundamentals, following the pay-as-you-go pricing model
- **Compute:**
 - Pay for compute time
- **Storage:**
 - Pay for data stored in the Cloud
- **Data transfer OUT of the Cloud:**
 - Data transfer IN is free
- Solves the expensive issue of traditional IT

Any data goes through Cloud is free. This solves expensive issues of Traditional IT.
* We only pay exactly of what we need. So we have huge cost saving ahead of us.



→ 2002 launched with amazon.com

→ Till 2006 in America, then 2007 launched in Europe and this all platforms runs on AWS cloud. (Netflix, Dropbox, Airbnb, NASA).

AWS cloud Number facts : -

* In 2023 AWS has \$90 billion in annual revenue.

* AWS accounts for 31% of market in Q1 2024 (Microsoft is 2nd with 25%.)

* Pioneer and leader of AWS cloud market for 3rd consecutive year.

* Over 1,000,000 Active users.

AWS cloud use cases: —

* AWS Enables you to build sophisticated, Scalable applications.

* Applicable to diverse set of Industries

* Use cases include

Enterprise IT, Backup and Storage, Big data Analytics.

* Web Hosting, Mobile and Social Apps.
* Gaming

McDonalds, 21st century fox, Activision and Netflix. Uses AWS

AWS Global Infrastructure: —

* AWS Regions * AWS Availability Zones * AWS Data Centers * AWS Edge locations/ Points of Presence.

AWS has Regions all around the world.

Names can be US east-1, EU-west-3, Asia specific etc.

How to choose an AWS Region?

If you need to launch a new application, where should you do it?



- Compliance with data governance and legal requirements: data never leaves a region without your explicit permission
- Proximity to customers: reduced latency
- Available services within a Region: new services and new features aren't available in every Region
- Pricing: pricing varies region to region and is transparent in the service pricing page

AWS Availability zones: —

* Each region has many availability zones

(usually 3 min is 3, max is 6) Example: —

Sydney; — ap-southeast-2

ap-southeast-2a

ap-southeast-2b

ap-southeast-2c

Each availability zone is one or more discrete data centers with redundant power, networking and connectivity. They are connected with high bandwidth, ultra-low latency networking.

* Amazon has 400+ Points of Presence (400+ Edge locations & 10+ Regional Caches) in 90+ cities across 40+ countries.

Tour of the AWS Console

- AWS has Global Services:

- Identity and Access Management (IAM)
- Route 53 (DNS service)
- CloudFront (Content Delivery Network)
- WAF (Web Application Firewall)

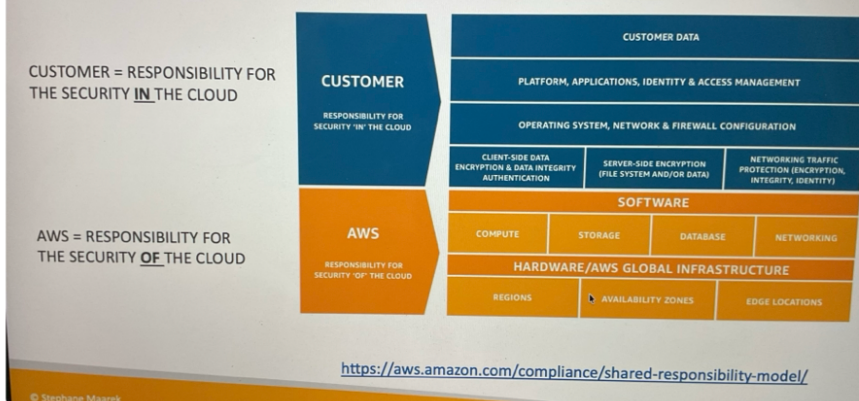


- Most AWS services are Region-scoped:

- Amazon EC2 (Infrastructure as a Service)
- Elastic Beanstalk (Platform as a Service)
- Lambda (Function as a Service)
- Rekognition (Software as a Service)



Shared Responsibility Model diagram



AWS Acceptable use policy: —

- * No illegal, harmful or offensive use or content
- * No security violations
- * No Network Abuse
- * No Email or other message abuse.

Quiz: — (what is cloud computing)

1) you only want to manage Application and data. what type of cloud computing model should you use?

- A) on premises b) Infrastructure as a Service (IaaS) c) Platform as a Service (PaaS)

d) Software as a Service (SaaS)

2) what is pricing model of cloud computing?

- A) Discount over time b) Pay as you go pricing c) pay once a year d) Flat rate pricing

3) which Global Infrastructure Identity is composed of one or more discrete data centers with redundant power, networking and connectivity, and are used to deploy infrastructure?

- a) Edge locations b) Availability zones c) Regions

4) which of following is NOT one of the 5 characteristics of cloud computing?

- A) Rapid elasticity and Scalability b) Multi tenancy and resource pooling c) Dedicated

Support Agent to help you develop applications d) on demand Self Service.

5) which are 3 pricing fundamentals of AWS cloud?

- A) Compute, Storage and Data transfer in AWS cloud b) Compute, Networking and Data transfer out of AWS cloud.
c) Compute, Storage and Data transfer out of AWS cloud d) Storage, function and Data transfer in AWS cloud.

6) which of following option is NOT a point of consideration when choosing an AWS Region?

- A) Compliance with data Governance b) latency c) Capacity Availability d) pricing

↳ Capacity is unlimited in cloud, you don't need to worry about it.

7) which of following is not an advantage of cloud computing? (you need to train employees more)

- a) Trade Capital expense (Capex) for operational expense (OpEx) b) Train your employee less
c) Go Global in minutes d) Stop spending money running and maintaining data centers.

8) AWS Regions are composed of?

- A) Two or more Edge locations b) one or more discrete data centers c) Three or more Availability zone

9) which of following has global scope?

- a) EC2 b) IAM c) lambda d) Rekognition

10) which of following is definition of cloud computing?

- a) rapidly develop, test and launch Software applications
b) Change Resource type when needed (on demand availability of computer system resource, especially data Storage and Computing power, without direct active management by user)
c) Automatic and quick ability to acquire resources as need them and release resources when no longer needed them.

11) what defines distribution of responsibility for Security in AWS cloud?

- A) AWS pricing fundamentals b) The Shared Responsibility model c) AWS Acceptance policy
d) AWS management Console

↳ who is responsible for what in AWS cloud.

12) A Company would like to benefit from advantage of public cloud but would like to keep sensitive assets in its own infrastructure. which deployment model should Company use?

...ing ...

a) Private cloud b) public cloud ~~c) Hybrid cloud~~ .

Using Hybrid cloud deployment model allows you benefit from flexibility, Scalability and demand storage access while keeping security and performance of your own Infrastructure.

13) What is Not authorized to do on AWS according to AWS acceptable use policy?

A) Building a gaming application d) Deploying a website ~~e) Run analytics on stolen Content~~

d) Backup your data

↳ You can run Analytics on AWS, but you cannot run on fraudulent Content.